

MPLAD-GUARD AI

Comprehensive Machine Learning & Artificial Intelligence Architecture Report

Executive Summary

MPLAD-GUARD AI integrates a multi-layered artificial intelligence engine designed to detect fraud, cost escalation, timeline overruns, agency concentration risks, and spatial/semantic work duplication in the Member of Parliament Local Area Development (MPLADS) Scheme. The pipeline combines unsupervised anomaly detection, natural language processing (NLP), spatial clustering, multi-criteria risk decomposition, graph neural relationship analytics, and retrieval-augmented generation (RAG) powered by LLMs.

1. Unsupervised Anomaly Detection (Isolation Forest)

Algorithm: Isolation Forest (via scikit-learn)

Source Code: [backend/pipeline/ml_anomaly.py](#)

Technical Description: The anomaly detection module partitions data spaces using random decision trees to isolate multivariate outliers without requiring labeled historical fraud data.

• **Input Feature Matrix (8 Dimensions):**

- **cost_z_score:** Standardized deviation of project cost against category peer median.
 - **cost_to_peer_ratio:** Ratio of sanctioned cost to median category cost benchmark.
 - **expenditure_ratio:** Ratio of disbursed expenditure against total sanctioned fund.
 - **delay_z_score:** Standardized execution delay in days beyond scheduled completion.
 - **delay_ratio:** Proportion of delay duration relative to original planned timeline.
 - **agency_delay_rate:** Historical proportion of delayed projects assigned to the implementing agency.
 - **projects_within_500m:** Count of active or completed MPLAD works within a 500-meter radius.
 - **max_semantic_similarity:** Maximum cosine similarity score against all existing work descriptions.
- **Configuration & Normalization:** n_estimators=100, contamination=0.12. Raw decision scores are Min-Max normalized into a normalized range [0.0, 1.0] where 1.0 represents the highest anomaly signal.

2. Natural Language Processing & Work Scope Duplication

Algorithms: TF-IDF Vectorization, Cosine Similarity, Regex Rule-Based Entity Resolution

Source Code: [backend/pipeline/feature_engineering.py](#) & [backend/pipeline/entity_resolution.py](#)

- **TF-IDF & Cosine Similarity:** Converts combined project names and detailed scope descriptions into TF-IDF n-gram vectors (n-grams 1–2, 500 max features). Computes pairwise Cosine Similarity matrix: $\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$.
- **Geographic Boost Logic:** If two projects share the same work type and block/ward location, a proximity boost (+0.45 for same ward, +0.20 for same block) is applied to detect disguised scope duplication.
- **Entity Resolution:** Resolves noisy agency names (e.g. 'rwd nalanda', 'r.w.d. nalanda') to canonical agency identities (e.g. AGY-RWD-NAL) using string normalization and alias dictionary matching.

3. Geospatial & Spatial Proximity Clustering

Algorithm: Haversine Great-Circle Distance & Radius Proximity Search

Source Code: [backend/pipeline/feature_engineering.py](#)

• **Haversine Distance Formula:**

$$d = 2R \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta \phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta \lambda}{2}\right)}\right)$$

- **Spatial Triggers:** Computes density counts for 500m and 2km radii. Flags an active **Spatial Duplication Risk** if a project of the exact same work type is located within **350 meters**.

4. Statistical Benchmarking & Outlier Analysis

Source Code: [backend/pipeline/feature_engineering.py](#)

- **Peer Group Normalization:** Groups project records dynamically by work_type (e.g. PCC Road, Community Hall, High-Mast Solar Light, Borewell).
- **Cost Z-Score:** $Z_{\text{cost}} = \frac{X_{\text{cost}} - \mu_{\text{peer}}}{\sigma_{\text{peer}}}$ to flag statistical financial inflation.
- **Timeline Z-Score:** $Z_{\text{delay}} = \frac{X_{\text{delay}} - \mu_{\text{delay}}}{\sigma_{\text{delay}}}$ to identify abnormal execution delays.

5. Multi-Criteria Explainable Risk Engine

Source Code: [backend/pipeline/risk_calculator.py](#)

The system computes 5 sub-risk dimensions (0–100 scale) and synthesizes an **Investigation Priority Score**:

- **Financial Risk (25%):** Evaluates cost-to-peer ratio (>2.2x triggers score >85) and cost Z-scores.
- **Timeline Overrun Risk (25%):** Evaluates absolute delay days (>300 days triggers score >85) and timeline overrun ratios.
- **Agency Risk (20%):** Profiles implementing agency's historical delay rate (>70% delay rate triggers High Risk).
- **Geographic Risk (15%):** Triggered by spatial proximity overlap (<250m distance triggers score >80).
- **Similarity Risk (15%):** Evaluates text description overlap (>85% similarity triggers score >85).

Composite Priority Equation:

$$\text{Priority Score} = 0.25 \times \text{Fin} + 0.25 \times \text{Time} + 0.20 \times \text{Agency} + 0.15 \times \text{Geo} + 0.15 \times \text{Sim}$$

6. Graph Intelligence & Relationship Analytics

Algorithm: NetworkX MultiDiGraph & Multi-Hop Traversal

Source Code: [backend/app/services/graph_service.py](#)

Constructs knowledge graph topologies connecting **Projects**, **Agencies**, **Locations**, and **Peer Projects**. Uncovers hidden collusion networks, agency concentration risks, and shared contractors.

7. Generative AI & Retrieval-Augmented Generation (RAG)

Model: Google Gemini API (gemini-1.5-flash / gemini-2.0-flash)

Source Code: [backend/app/services/rag_service.py](#)

Incorporate MoSPI MPLADS Scheme regulatory guidelines (Para 4.12 on 5-year duplication caps, Para 3.4 on State Schedule of Rates, Para 6.1 on Measurement Book verification) into LLM context prompts to generate structured investigation briefs.

8. Master Summary of Algorithms

Algorithm / Technique	Library / Module	Input Data	Output Signal	File Location
Isolation Forest	scikit-learn	8 numerical feature dimensions	Anomaly Score (0.0-1.0)	ml_anomaly.py
TF-IDF Vectorizer	scikit-learn	Project titles & descriptions	500-dim feature vectors	feature_engineering.py
Cosine Similarity	scikit-learn	TF-IDF text vectors	Similarity Score (0.0-1.0)	feature_engineering.py
Haversine Distance	Math / Custom	Latitude & Longitude coordinates	Proximity Distance (km)	feature_engineering.py
Z-Score Standardization	NumPy / Pandas	Cost & Delay benchmarks	Z-Score (+/- Std Dev)	feature_engineering.py
Risk Decomposition Engine	Custom Rules Engine	Sub-risk dimension scores	Priority Score (0-100)	risk_calculator.py

NetworkX MultiDiGraph	NetworkX	Entities & Relationship Links	Graph Topology Nodes/Edges	graph_service.py
RAG + Gemini LLM	google-generativeai	DB Context & MoSPI Rules	Investigation Audit Brief	rag_service.py

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